The HRBuddy chatbot consists of 3 main parts:

(1) Frontend

(2) Backend
(3) Database

Frontend

- There must always be a singular frontend file, called ‘index.html’.
- Even if multiple files are created within the frontend folder, only one page is responsible for the frontend UI, which is ALWAYS to be titled ‘index.html’.
- Naming the wrong file ‘index.html’ will result in the wrong/not working frontend display, so please name accordingly.

Server procedure: Any changes made to the ‘index.html’ file will only reflect if the changes are done on the machine that hosts the server that the chatbot is running on.

All changes to be made (if any), are to be compiled, the code must be re-written as necessary, and the same must be sent to the IT department, who will then replace the old ‘index.html’ file with the newly sent file.

Backend

- The backend file can be called anything, but for convention purposes, the file is currently called ‘main.py’.
- Similar to the frontend, the existence of only a single backend file is recommended, irrespective of the name.

Server procedure: Same as frontend.

Database

- The database and its properties are designed and controlled by the backend ‘main.py’ file.
- The database is created when the chatbot is started up in the new server for the very first time and then exists perpetually from then on.
- All database-related procedures with respect to addition and deletion of files are listed in a separate document.
- The database folder is called ‘Chroma’ within the program files of HRBuddy. If any database issues are encountered, or a database cleansing is necessary, please delete the entire Chroma folder and then re-start/start-up the chatbot again. The database is automatically re-built using the documents stored in the documents folder.

Documents

The documents folder is critical, but straightforward. All the documents/information that you want the chatbot to be familiar with, must be uploaded here in PDF form (ONLY, ideally). These documents are indexed by the backend and stored as the Chroma database, that the chatbot references to answer queries.

There is no official format in which the PDF doc to be uploaded must be written, the only guidelines are:

- The document must be in clear, concise English.
- The document must not have contradicting information, within itself or with other documents.
- The document must not deviate from HR-related information.
- If similar terms, keywords or names are used in multiple, distinct contexts, those same contexts must be clearly defined along, to avoid confusion.

Server procedure: Same as frontend.

Steps to set up the chatbot:

(1) Download and install Python and Ollama (check installations folder).

(2) Open Command Prompt and pull a model. Instructions and model names are specified in the 'LLM Models' document in the Google Drive.

(3) Open command prompt as administrator and create your project folder.

Open **Command Prompt** and run:

```
mkdir C:\hrbuddy
cd C:\hrbuddy
mkdir documents
mkdir backend
```

Place all your HR documents (PDFs, DOCX files, plain text files) into `C:\hrbuddy\documents\`. These are your pay structure policy, rules, leave policy, uniform policy — anything the chatbot should know.

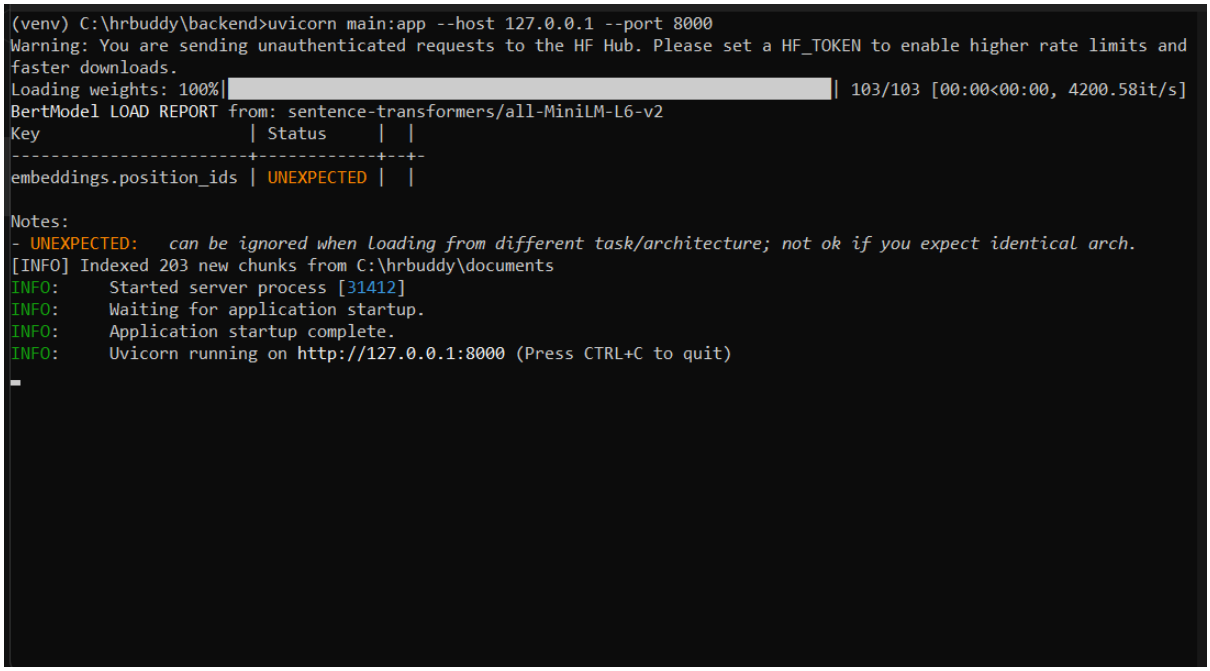
Note: Also carry out 'mkdir frontend' immediately after 'mkdir backend'.

(4) In the same prompt, download and install the requirements document (in the Google Drive, commands included within).

(5) Open your file explorer, and within the frontend folder inside the HRBuddy folder, copy and paste the 'index.html' file (from the Google Drive). Do the same for the 'main.py' file, but in the backend folder inside the HRBuddy folder.

(6) Start the backend (server start-up commands mentioned in a separate document).

(7) Wait until the below lines show up on the screen. If successful, these are the lines that will show up. Once the final line shows, please wait anywhere between 1-3 minutes, before attempting to open the chatbot.



(8) Type the link in your browser (in this example: http://127.0.0.1:8000, because it is being run on a laptop/desktop, but when running it on Marriott server, the URL depends entirely on the IT department – separate document included in the Google Drive regarding server procedures).

Notes:

(1) In the screenshot shared above, there is a line that says Indexed 203 new chunks. This only happens if either the Chroma database has been completely deleted, or if a new document has been added to the Documents folder inside HRBuddy folder. Otherwise, for regular start-up, it will say ‘Indexed 0 new chunks’.

(2) The words ‘server’ and ‘backend’ have been used interchangeably when talking about running the chatbot on a personal computer or a desktop. The two terms only differ when IT department is involved, with respect to running the chatbot on the Marriott server.